

DRIVERLESS VEHICLE

EDGE-CASE INTELLIGENCE PLATFORM

Turning risky dashcam footage into structured, searchable and explainable incident records for autonomous-driving development, validation and safety review.



RARE-EVENT DISCOVERY

Automatically mine and surface safety-critical events from long-form dashcam video.



MULTIMODAL CLASSIFICATION

Fuse motion, object, tracking and context evidence to classify near_miss vs collision with high accuracy.



QWEN-BASED SEMANTIC UNDERSTANDING

Qwen2.5-VL (LoRA + SFT) generates tags, summaries and timelines with schema-constrained outputs.



SEARCHABLE INCIDENT RECORDS

Turn outcomes and semantics into structured, searchable records for review, comparison and hard-case mining.



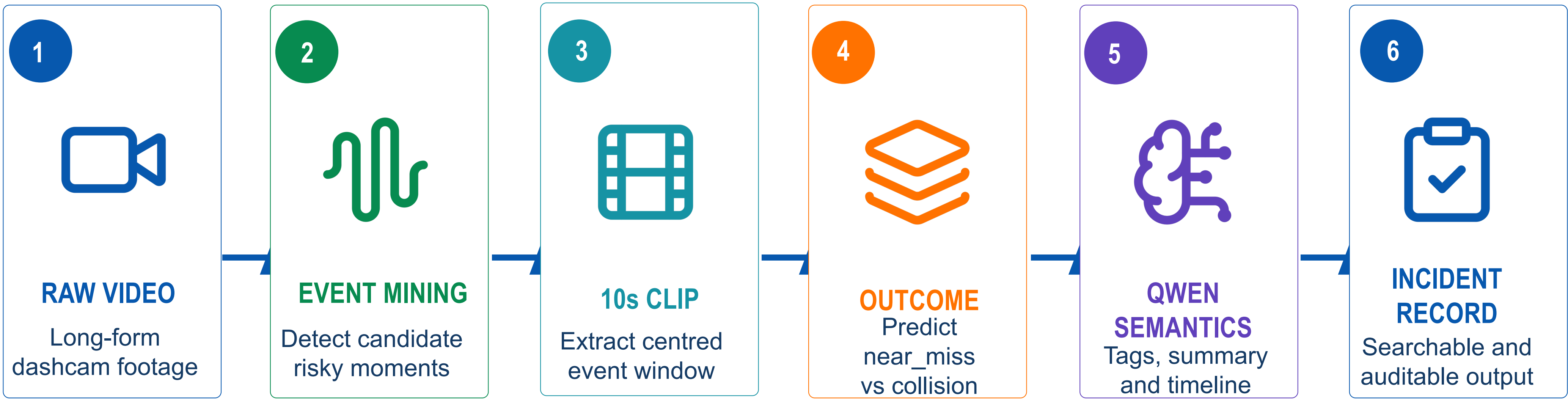
CHALLENGE

RARE EVENTS.
HIGH STAKES.

Rare but safety-critical edge cases are deeply buried in large dashcam datasets. Manual review is slow, expensive and inconsistent. Many existing datasets also collapse near-misses and collisions into one broad “risky” category, losing important distinctions needed for failure analysis, scenario coverage and safety evaluation.

We therefore need a pipeline that not only classifies event outcome, but also generates human-readable incident understanding.

HOW IT WORKS



The platform converts long-form dashcam footage into standardised incident records. It first mines candidate risky moments from raw video, then extracts 10-second event-centred clips so that each sample contains the build-up, critical interaction and short aftermath of the event. The extracted clip is passed through an outcome classifier to predict whether the case is a near_miss or collision, while Qwen2.5-VL generates structured semantic understanding such as tags, summary and timeline. Finally, the predicted outcome, confidence score and semantic fields are merged into one searchable incident record for review, comparison and dataset curation.

OUR APPROACH

A

UPSTREAM DETECTION + OUTCOME CLASSIFICATION

An upstream model scans long-form footage to identify candidate risky windows. A multimodal classifier then predicts near_miss vs collision. A correction/refinement model improves ambiguous or noisy cases.

- Technical evidence sources:
- Camera motion & video features capture sudden shake, braking and angle-related patterns.
- YOLO-based object residual features encode object interactions and local visual cues.
- CoTracker dynamics model how agents move and interact over time.
- Long-context probability anchors provide broader temporal evidence around the event.
- LightGBM late-fusion combines all signals in a robust, data-driven fusion strategy.

B

QWEN-BASED SEMANTIC UNDERSTANDING

Qwen2.5-VL (post-trained with LoRA and SFT) analyses the 10-second clip to produce structured, human-readable incident understanding.

- Scenario tags (e.g., pedestrian_crossing, cut_in, red_light).
- Concise natural-language summary of what happened.
- Timeline-style description of key events.
- Actors (vehicles, pedestrians, cyclists, others).
- Environment (weather, lighting, road type, scene context).
- Evidence: object interactions, motion, tracking, signal state.
- Event progression with causal/temporal ordering.
- Schema-constrained output for consistency, searchability and analytics.

C

STRUCTURED INCIDENT CARD + PLATFORM LAYER

The final output is a reusable incident record—not just a label. It includes outcome, confidence and rich semantic context.

- Key fields: clip_id, source_dataset, outcome, confidence, tags, actors, environment, timeline, summary, review_status, label.
- Enables scalable searchable review and case comparison.
- Supports dataset curation, scenario coverage analysis and hard-case mining.
- Human-in-the-loop review keeps records auditable, correctable and useful for downstream evaluation.

EVIDENCE

STRUCTURED INCIDENT CARD EXAMPLE

clip_id	CLIP_2024_05_18_00123	timeline	–5s Approach –3s Interaction 0s Event +2s Evasion +5s Clear
status	completed	processing_seconds	9.3566s
label	near_miss	behaviour	braking
confidence	medium	summary	Vehicle turning right yields to pedestrian crossing. No impact.
tags	pedestrian_crossing, right_turn, yielding, urban, no_impact		
actors	vehicle, pedestrian		
environment	daylight, clear, urban, dry		

All outputs follow a schema-constrained format for consistency and searchability.

KEY RESULTS



93.96%
TEST ACCURACY



0.965
TEST AUROC



0.939
MACRO F1

- ✓ Evaluated on a fixed Nexar test split.
- ✓ 744 cleaned and labelled positive clips, balanced between near-miss and collision.
- ✓ 4063-dimensional features drawn from motion, object, tracking and long-context signals.
- ✓ SAVED data prepared for external validation, hard-case mining and richer semantic tagging.
- ✓ Compared with our earlier BADAS1.0-based pipeline, AUROC improved from 0.85 to 0.965 under our project evaluation setup.

TECHNICAL CONTRIBUTIONS



SEPARATE CLASSIFICATION FROM EXPLANATION

Outcome classification is decoupled from semantic understanding to improve accuracy and interpretability.



UPSTREAM + CORRECTION + QWEN INTEGRATION

Combine upstream detection, multimodal classification, correction models and Qwen for robust results.



FROM CLIPS TO STRUCTURED SEARCHABLE RECORDS

Transform isolated 10-second clips into standardised incident cards for scalable review and analytics.



BUILT FOR SAFETY VALIDATION AND ITERATION

Enables dataset curation, scenario coverage analysis and continuous improvement loops.